5.6 The Fermat Point

WORKSHEET

The question

STEP III 2000, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Where are A, B and C? What are their arguments? What angles do OA, OB and OC make with each other?

STAGE 1 The key identity

Covers: the identity

Ask yourself What is α^3? Factorise $z^3 + 1$, and use $\alpha \neq -1$.

Ask yourself Where are A, B and C? Find their arguments.

STAGE 2 Equilateral triangles as rotations

Covers: the vertices N, L and M

Ask yourself Triangle CAN is equilateral, lettered clockwise. Standing at C, how do you get from A to N? Write $N - C$ in terms of $A - C$.

Pause and try Rotating by 60° clockwise is multiplying by $e^{-i\pi/3}$. Write that in terms of α, then find L and M in the same way and simplify with the identity.

STAGE 3 Three lines through the origin

Covers: the three lines

> **Pause and try** Compare L with C: both are real multiples of the same complex number. What does the sign of each multiple tell you? Do the same for M and A, and for N and B.

What's really going on

Why the Fermat point minimises distance

Why should $PA + PB + PC$ be smallest at the point where the sides subtend $120°$? Here is a lovely argument. Rotate the whole picture about B by $60°$, sending A to L' (the apex of an equilateral triangle on AB) and P to P'. Triangle BPP' is equilateral, so $PB = PP'$, and $PA = P'L'$. Then $PA + PB + PC = L'P' + P'P + PC$, the length of a path from L' to C. That is at least $|L'C|$, with equality exactly when the path is straight, which happens when the angles at P are $120°$. So the minimum total length is $|LC| = p + q + r$, as found in the question. (This works provided every angle of the triangle is less than $120°$; otherwise the minimum is at the obtuse vertex.)

Soap films and Steiner trees

Dip a frame with three pegs (representing A, B, C) into soapy water: the soap film, minimising its surface tension energy, forms three walls meeting at $120°$ at the Fermat point. For more points the shortest connecting network is a *Steiner tree*, with extra junctions where three edges meet at $120°$. Finding Steiner trees is computationally hard in general (it is an NP-hard problem), but they matter in practice for designing networks of roads, pipelines and the wiring on computer chips.

Napoleon's theorem

The same configuration contains another gem: the *centres* of the three equilateral triangles on the sides of any triangle form an equilateral triangle. This is traditionally credited to Napoleon Bonaparte (who was a keen amateur mathematician), though there is no evidence he discovered it. It too has a short proof with complex numbers,

using $1 + \alpha^2 = \alpha$.

Try this next

1. **Check the lettering.** Show that if the equilateral triangles were lettered anticlockwise (built inwards), the three lines would generally not pass through O.

2. **Napoleon's theorem.** Show that the centre of the equilateral triangle ABL is $\frac{1}{3}(A + B + L)$. Using the expressions above, show that the three centres form an equilateral triangle.

3. **The minimum.** With $p = 2$, $q = 1.4$, $r = 1.1$, compute $PA + PB + PC$ for P at each vertex and at the centroid $\frac{1}{3}(A + B + C)$, and check that all exceed 4.5.

4. **An equilateral test.** Show that z_1, z_2, z_3 form an equilateral triangle (in either orientation) if and only if $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$.

Further reading

- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 7 discusses the Fermat point, Steiner trees and soap films.
- **Liang-shin Hahn, *Complex Numbers and Geometry*** (Mathematical Association of America). Many geometric theorems proved with complex numbers, at exactly this level.
- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 1 includes Napoleon's theorem and other geometric results.

The full guided solution is in Section 5.6:
https://beyondstep.keithmcnulty.org/ch05/fermat-point.html